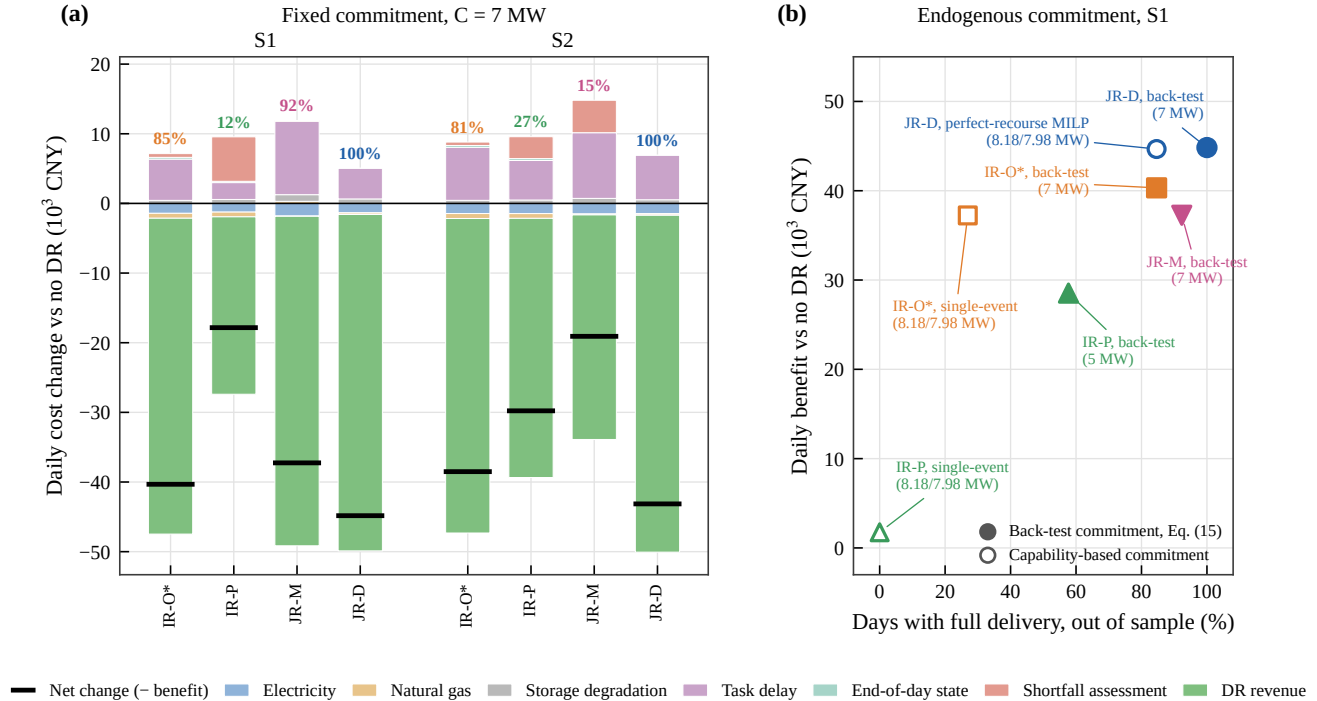
(a)
Fixed commitment, C = 7 MW
S1
S2
Daily cost change vs no DR (10^3 CNY)
20
10
0
−10
−20
−30
−40
−50
85%
12%
92%
100%
81%
27%
15%
100%
IR-O*
IR-P
IR-M
JR-D
IR-O*
IR-P
IR-M
JR-D
(b)
Endogenous commitment, S1
Daily benefit vs no DR (10^3 CNY)
50
40
30
20
10
0
JR-D, back-test
(7 MW)
JR-D, perfect-recourse MILP
(8.18/7.98 MW)
IR-O*, back-test
(7 MW)
JR-M, back-test
(7 MW)
IR-O*, single-event
(8.18/7.98 MW)
IR-P, back-test
(5 MW)
IR-P, single-event
(8.18/7.98 MW)
Back-test commitment, Eq. (15)
Capability-based commitment
0
20
40
60
80
100
Days with full delivery, out of sample (%)
Net change (− benefit)
Electricity
Natural gas
Storage degradation
Task delay
End-of-day state
Shortfall assessment
DR revenue